Supplemental Material: Reliability-aware active learning for FFLO phase spaces

Author list pending human approval

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SUPPLEMENTAL MATERIAL

PHYSICAL MODEL AND NOTATION

The Hamiltonian, Nambu convention, and parameter definitions are identical to those in the main text and the canonical model specification. The matrix convention is row = J_A and column = k_BT for two-dimensional arrays. No thermodynamic phase definition is changed by the active-learning or topology layers.

EXACT BDG ORACLE AND RELIABILITY HIERARCHY

At fixed external parameters, the oracle scans the configured q and Δ domains, refines candidate basins, and compares the best superconducting free energy with the normal reference. A q-edge optimum is not accepted until its window is expanded or it is explicitly marked unresolved. Near the Normal–SC boundary, the strictly positive-Δ gap $\min_{\Delta\geq\Delta_\epsilon} F(\Delta, q) - F(0, q)$ distinguishes trusted normal points from a finite-resolution boundary band. Reliability fields are stored independently of phase labels.

For topology, the analytic Pfaffians in the project convention are

$$P_0 = (\mu - t\cos(q/2))^2 + \Delta^2 - (\lambda_y \sin(q/2))^2 - (J_A \cos(q/2) + \lambda_z \sin(q/2))^2, \tag{1}$$

$$P_\pi = (\mu + t\cos(q/2))^2 + \Delta^2 - (\lambda_y \sin(q/2))^2 - (J_A \cos(q/2) + \lambda_z \sin(q/2))^2. \tag{2}$$

Away from the Pfaffian tolerance, $P_0P_\pi < 0$ gives $\nu = 1$ and $P_0P_\pi > 0$ gives $\nu = 0$. The invariant is accepted only together with a full-Brillouin-zone gap exceeding its scale-aware tolerance and trusted thermodynamics. Direct analytic/numeric Pfaffian comparison agreed for 104/104 validation points, and the CPU/GPU bulk-gap discrepancy was 7.77×10^{-16} in the archived validation.

ACTIVE-LEARNING LOOP AND ACQUISITION

Each restartable iteration trains five-member regression/classification ensembles, scores a fixed candidate domain, excludes existing coordinates, applies soft observation and batch repulsion, evaluates the selected microbatch with the exact oracle, and appends only accepted records. The two-dimensional production configuration used a 512-call random-grid cold start, batches up to 256 points, and a 15% validation split. The acquisition score combines classification entropy/margin, phase-facing regression uncertainty, Normal–SC and uniform-SC–FFLO proximity, gated q-edge risk, and conservative extrapolation. Response observables are retained as diagnostics and do not replace thermodynamic boundary targets.

THERMODYNAMIC CONVERGENCE AND HARD-RISK AUDIT

The Stage-II endpoint contains 7,434 observations. Its final phase-map change is 0.0016287, boundary coverage p_{95} is 0.0046875, and trusted surprise is 0/127. The 129-call hard-risk layer contains 110 rerun-required and 19 other untrusted or training-ineligible records. Counterfactual boundary-impact tests found no cluster requiring a targeted rerun under the adopted publication tolerance; the layer remains visible and is not declared eliminated.

TOPOLOGY CONVERGENCE AND SENSITIVITY

The formal Stage-III audit uses trusted gapped FFLO points, deterministic k-nearest-neighbor inverse-distance interpolation of P_0 and P_π on a 401×401 monitor grid, and support radius 0.045 in normalized coordinates. With $k = 8$, the final coverage is $p_{95} = 0.006185$ below the 0.00625 threshold; the last-three map changes, boundary shifts, and trusted surprises pass their registered thresholds and retain one principal component. The $k = 6$ sensitivity coverage 0.006269 is slightly above threshold, whereas $k = 12$ gives 0.005913 and two significant components. Local opposite-label brackets are support diagnostics, not exact boundaries.

STAGE-IV FINITE-DOMAIN DIAGNOSTICS

The three-dimensional run samples $(k_BT/t, J_A/t, \mu/t)$ over the finite interval $\mu/t \in [-0.5, 1.5]$ and returns 7,081 unique exact observations. The main-text figure reports the exact thermodynamic cloud, trusted FFLO topology support, and four explicit fixed-μ bins. These are discrete observations; no continuous topology surface is fitted or inferred.

Figure S1 records archived numerical histories. The map-change fraction measures changes between successive proxy label maps; the proxy-surface displacement is a normalized bidirectional p_{95} distance; and trusted surprise is the fraction of new trusted labels that differ from the previous proxy prediction. These quantities test stability of a particular sampling and reconstruction procedure. They are not thermodynamic observables or universal physical scales. We therefore show their raw values without a cutoff line and do not convert them into a physical convergence declaration. The archive additionally records one final proxy-coverage value, 0.015318, and component counts $3 \to 4 \to 3$ over the last three iterations; neither supplies a standalone physical closure criterion.

EXPLORATORY ACQUISITION AND WORLD-MODEL LIMITATIONS

The matched V3/V4 analysis is an offline comparison of frozen runs. The two methods trade support density against sequential surprise, and V4 does not improve the shared graph-uncertainty metric or validate any of four candidate nodes. The Stage-VI hierarchy improves field masking but lacks the spectral artifacts required for a physical spectral descriptor. These negative and incomplete results constrain the Discussion and Outlook; they are not promoted to Methods claims.

AI-ASSISTED WORKFLOW AND REPRODUCIBILITY

The manuscript pipeline uses a versioned claim/evidence registry, figure contracts, artifact hashes, machine-readable paragraph traces, and explicit human gates. AI-assisted tools organize, check, and render registered evidence. They do not authorize new physical claims, alter frozen data, or replace human scientific approval. This review package is generated by the same checkpointed LangGraph thread that produced the approved Results package.

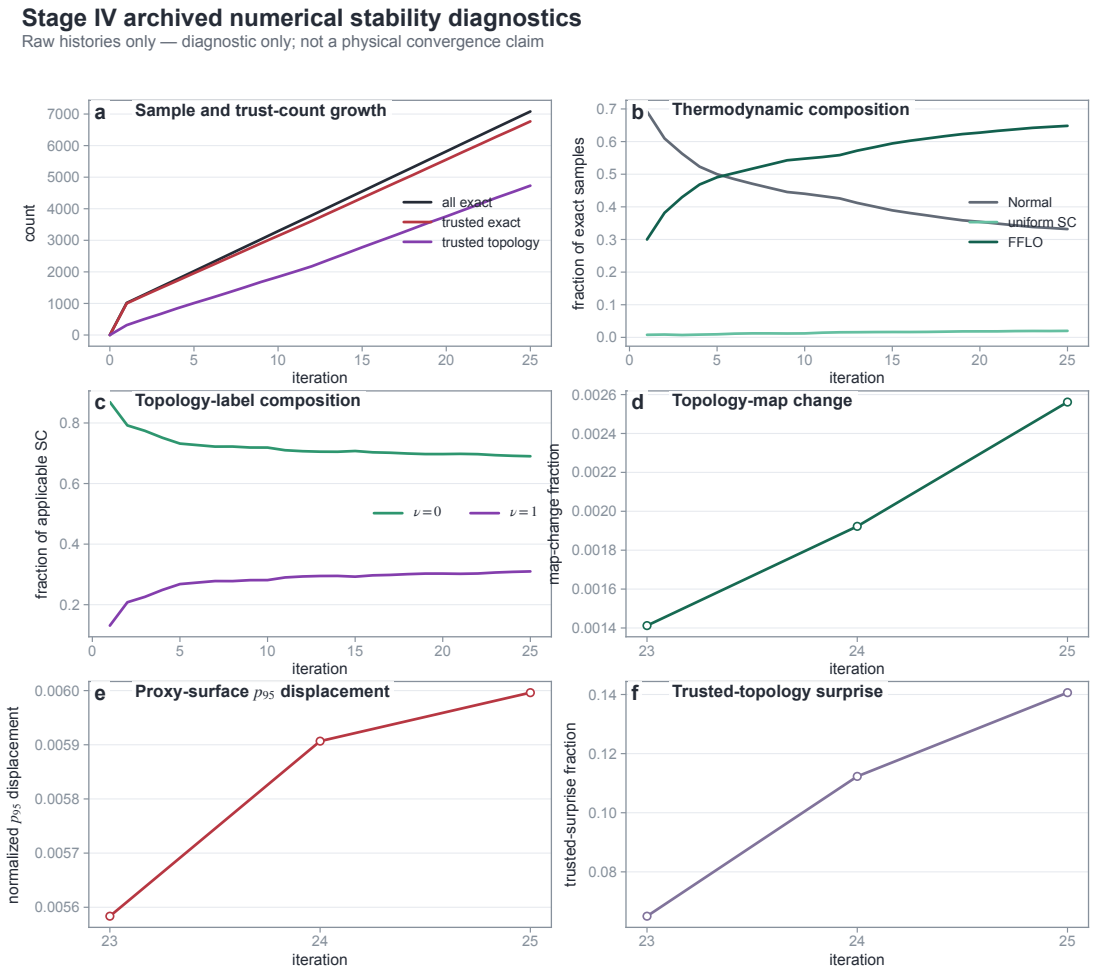


FIG. S1. **Archived Stage-IV numerical stability diagnostics.** **(a)** Total exact, trusted-exact, and trusted-topology counts. **(b)** Thermodynamic-label fractions and **(c)** applicable-SC topology-label fractions versus iteration. **(d)**–**(f)** Last-three raw histories of topology-map change, proxy-surface p_{95} displacement, and trusted-topology surprise. No threshold line is drawn and no panel is used as a physical convergence criterion.